Lab 3 – Architectural Justification

Self-Service Coffee Kiosk System

Architecture Selection: We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

1. Architectural Choice

The selected architectural style is Layered Architecture, organised into three logical layers. The **Presentation Layer** contains the User Interface Component (touchscreen). The **Business Layer** contains the Order Manager, Payment Service and Receipt Printer Components. The **Data Layer** contains the Menu Database Component. Each layer communicates with the next only through the defined interfaces shown in the component diagram.

2. Two Reasons for Choosing the Architecture

Reason 1 – Clear separation of concerns. The kiosk has distinct responsibilities: handling touchscreen interactions, selecting a coffee (Espresso, Americano or Latte) in a size (Small or Large), processing credit-card payment, printing the receipt, and storing menu and pricing data. Layered Architecture maps each responsibility to a specific layer and component, so the screen design can be changed in the User Interface Component without affecting the order, payment or pricing logic.

Reason 2 – Hardware and data are isolated behind interfaces. The kiosk must connect to receipt printer hardware and must store menu data and pricing information. In this architecture the Receipt Printer Component and the Menu Database Component are accessed by the Order Manager only through the Receipt Interface and the Database Interface. Because each component is a replaceable black box, the printer or the menu storage can be changed without modifying the Order Manager Component.

3. Security Advantage

The kiosk accepts credit-card payment only, so payment data must be protected. In Layered Architecture the touchscreen (Presentation Layer) never communicates directly with the Payment Service Component or the Menu Database Component. All requests pass through the Order Manager Component, which forwards payment requests through the Payment Interface and receives only the payment success or failure status in return. Card handling is therefore restricted to one component in the Business Layer, and the menu and pricing data in the Data Layer cannot be altered directly from the customer-facing screen.

4. Performance Benefit

The kiosk is a single, self-contained ordering system with a small and fixed menu of three coffee types and two sizes. Layered Architecture keeps the interaction path short and predictable: the User Interface Component sends one order request to the Order Manager Component, which retrieves menu and pricing information with a single query through the Database Interface. Since the layers communicate through direct interface calls rather than over a network, the customer receives quick responses to each touchscreen selection and the receipt is printed immediately after payment succeeds.